

8	(a) photon energy = hc/λ $= (6.63 \times 10^{-34} \times 3.0 \times 10^8)/(590 \times 10^{-9})$ $= 3.37 \times 10^{-19} \text{ J}$	C1 C1	
	number = $(3.2 \times 10^{-3})/(3.37 \times 10^{-19})$ $= 9.5 \times 10^{15}$ (allow 9.4×10^{15})	A1	[3]
	(b) (i) $p = h/\lambda$ $= (6.63 \times 10^{-34})/(590 \times 10^{-9})$ $= 1.12 \times 10^{-27} \text{ kg m s}^{-1}$	C1 C1	
	total momentum = $9.5 \times 10^{15} \times 1.12 \times 10^{-27}$ $= 1.06 \times 10^{-11} \text{ kg m s}^{-1}$	A1	[3]
	(ii) force = $1.06 \times 10^{-11} \text{ N}$	A1	[1]